



Project Report

Automated Face Recognition Attendance System

Course Details

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1. Executive Summary

AttendX is a modern, end-to-end web application that revolutionizes educational attendance tracking by replacing manual roll-calls and basic sign-in sheets with AI-powered facial recognition and dynamic session verification.

Built with a high-performance Python/Flask backend, a responsive Vanilla CSS/JS frontend, and a cloud-native Supabase (PostgreSQL) database, AttendX guarantees secure, tamper-proof attendance marking while delivering a sleek, dark-themed user experience for Students, Teachers, and Administrators.

2. Technology Stack

The system architecture connects a Vercel-hosted frontend, a Render-hosted Flask backend with an integrated face-recognition engine, and a Supabase PostgreSQL database, with Brevo handling transactional email delivery.

Layer	Technology	Key Purpose & Highlights
Frontend UI/UX	HTML5 & Vanilla CSS	Custom design system featuring dark mode, glassmorphic cards, smooth micro-animations, and responsive CSS Grid/Flexbox layouts. Zero heavy UI frameworks.
Client Logic	Vanilla JavaScript (ES6+)	Modular state management, Fetch API client (utils.js), real-time camera stream capture (getUserMedia WebRTC), and dynamic DOM rendering.
Backend API	Python 3 / Flask	Lightweight, high-throughput RESTful API handling authentication, role-based authorization, session management, and business logic.
AI / Biometrics	OpenCV & dlib Face Recognition	Extracts 128-dimensional facial embeddings from webcam captures, performing cosine similarity / Euclidean distance matching to verify student identities.
Database	Supabase (PostgreSQL 15)	Relational schema storing users, courses, enrollments, facial biometric vectors, attendance sessions, and timestamped attendance logs.
Email Service	Brevo HTTP API	Automated transactional email service using custom-designed Learnova-style HTML email templates for user onboarding and attendance notifications.
Cloud Deployment	Render & Vercel	Vercel delivers the static frontend globally via edge CDN; Render hosts the Python Flask backend containers with automatic CI/CD.

3. Value Proposition & Features

For Students

- **Biometric Identity Enrollment:** capture and register facial vectors securely using any standard laptop or mobile webcam.
- **Frictionless Attendance Marking:** verify identity against active classroom sessions in seconds, preventing buddy punching.
- **Live Attendance Dashboard:** real-time attendance percentages, course schedules, and historical records.
- **Instant Email Receipts:** styled confirmation emails upon registration and attendance events.

For Teachers & Instructors

- **Course & Classroom Management:** create and organize multiple academic courses with custom course codes and descriptions.
- **Dynamic Attendance Sessions:** launch time-bound sessions with optional QR codes or direct classroom verification.
- **Real-Time Roster Monitoring:** watch students check in live, view statistics, and identify chronic absences.
- **Manual Overrides & Export:** excused absence management and instant CSV export of attendance rosters.

For Administrators

- **System-Wide Analytics:** metrics tracking total active users, enrolled courses, and daily attendance volumes.
- **Role & User Management:** complete oversight over student, teacher, and administrator accounts.
- **Security & Audit Logs:** traceable records of all biometric enrollments and system actions.

4. Frontend Pages & Feature Catalog

The table below lists every page built into the AttendX platform.

Page Path	Target Role	Key Features & Functional Workflow
index.html	Public	Landing Page introducing the platform, highlighting AI biometric capabilities, with responsive navigation and sign in / create account options.
pages/login.html	All Users	Secure email/password login with role-aware redirection to Student, Teacher, or Admin dashboard.

pages/register.html	New Users	Multi-field registration supporting role selection (Student/Teacher) and automated welcome email dispatch via Brevo.
pages/student/dashboard.html	Student	Overall attendance percentage, enrolled courses, quick-action navigation, and recent activity logs.
pages/student/enroll.html	Student	Camera interface that captures a live photo, generates facial embeddings via the backend, and enrolls the student in new courses.
pages/student/mark.html	Student	Activates webcam stream, captures the student's face during an active session, and validates identity against stored embeddings.
pages/teacher/dashboard.html	Teacher	Summary of active courses, total student reach, quick course creation modal, and session status cards.
pages/teacher/course.html	Teacher	Launch attendance sessions, display session QR codes, track live check-ins, and download CSV reports.
pages/admin/dashboard.html	Admin	Organization-wide statistics, user directory management, and system health oversight.

5. End-to-End Biometric Verification Workflow

The verification sequence below shows how a student's attendance is captured, matched, and recorded in real time:

- Student opens “Mark Attendance” on the mark.html page and starts the webcam.
- The browser captures a live video frame and encodes it as base64 JPEG.
- The frame is POSTed to /api/attendance/mark along with the course_id and session_id.
- The Flask backend fetches the student's stored 128-dimensional facial embedding from Supabase.
- The face recognition engine compares the live image against the stored vector using cosine similarity / Euclidean distance.
- A match is confirmed when the distance is below the 0.60 threshold.
- The backend inserts a new attendance record with status “present” and a timestamp.
- A Learnova-style HTML receipt email is dispatched via the Brevo API.
- The frontend displays a green success checkmark and updates the student's attendance stats.

6. Project Highlights & Quality Benchmarks

- Zero-Dependency Modern CSS: highly customized aesthetics with smooth gradients (#6C63FF primary branding), glassmorphic card overlays, and subtle hover animations without bulky CSS frameworks.

- Resilient Cloud Networking: HTTP-based email APIs (requests + Brevo) and CORS-ready Flask configuration operate seamlessly across cloud restrictions on Render and Vercel.
- Security Best Practices: password hashing, JWT/token-based session headers, and server-side biometric vector evaluation ensure user privacy and academic integrity.

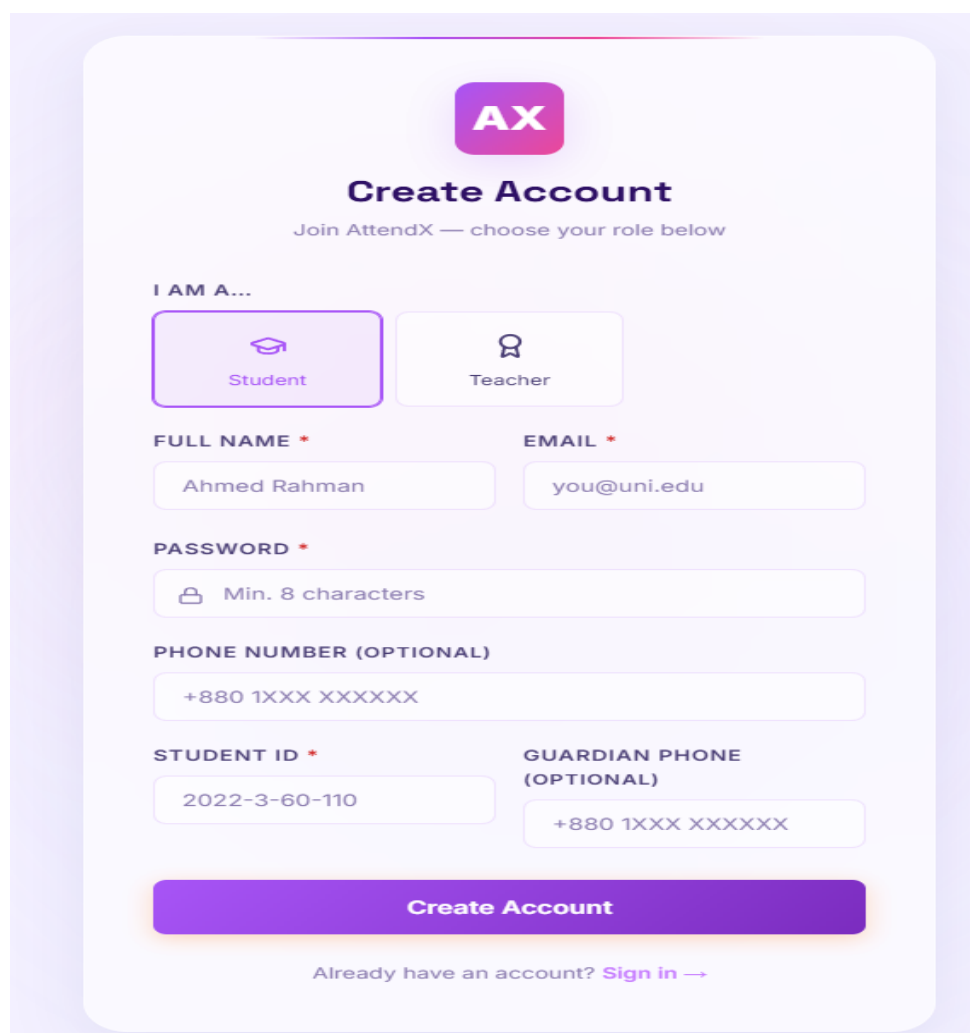
7. How to Use AttendX (User Manual)

This section walks each user role through AttendX step by step, from account creation to daily use. Screenshot placeholders are shown below; replace them with the actual application screenshots supplied separately.

7.1 Getting Started

Step 1: Visit the landing page Open the AttendX website and review the platform introduction and features.

Step 2: Create an account Click “Sign Up,” fill in your name, email, password, and select your role (Student or Teacher).



The image shows a registration form for AttendX. At the top is the 'AX' logo. Below it is the title 'Create Account' and the subtitle 'Join AttendX — choose your role below'. The form has two columns for role selection: 'Student' (with a graduation cap icon) and 'Teacher' (with a person icon). Below this are input fields for 'FULL NAME *' (containing 'Ahmed Rahman'), 'EMAIL *' (containing 'you@uni.edu'), 'PASSWORD *' (with a lock icon and 'Min. 8 characters'), 'PHONE NUMBER (OPTIONAL)' (containing '+880 1XXX XXXXXX'), 'STUDENT ID *' (containing '2022-3-60-110'), and 'GUARDIAN PHONE (OPTIONAL)' (containing '+880 1XXX XXXXXX'). A large blue 'Create Account' button is at the bottom, followed by a link 'Already have an account? Sign in →'.

Figure 7.1 — Registration Page

Step 3: Check your inbox — A welcome email confirms your registration.

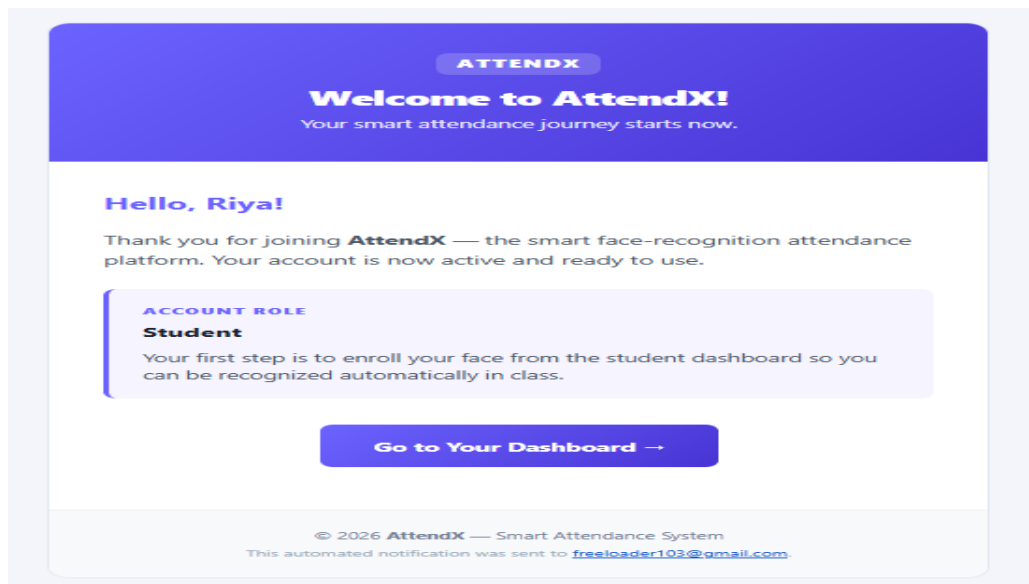


Figure 7.2 — Welcome Email

Step 4: Log in — Enter your email and password on the login page; you will be redirected to the dashboard for your role.

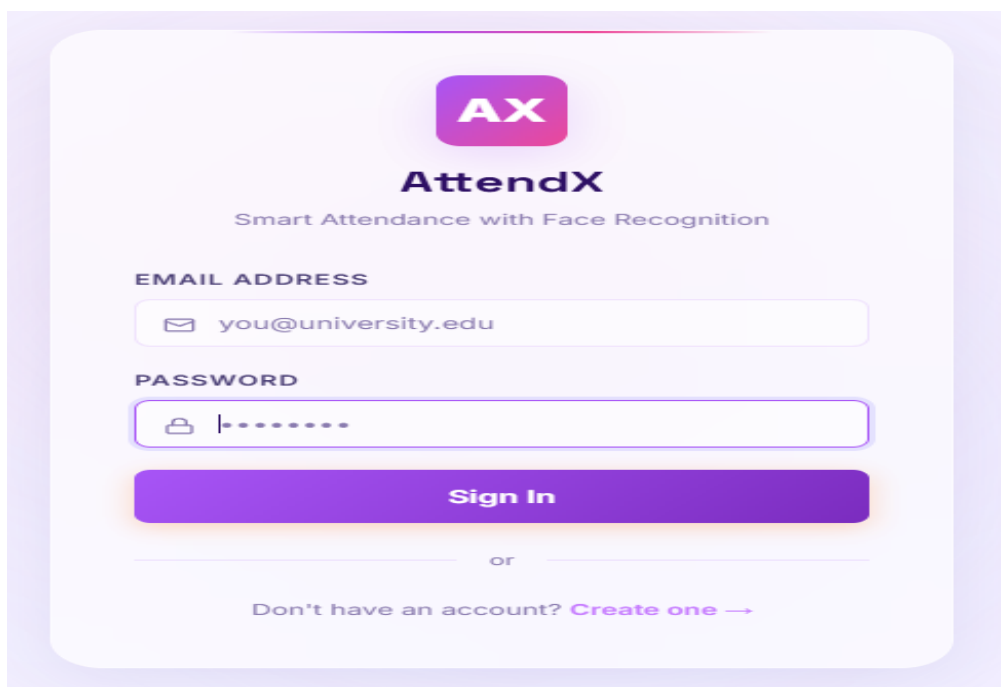


Figure 7.3 — Login Page

7.2 Student Manual

The dashboard shows your overall attendance percentage, enrolled courses, and recent activity.

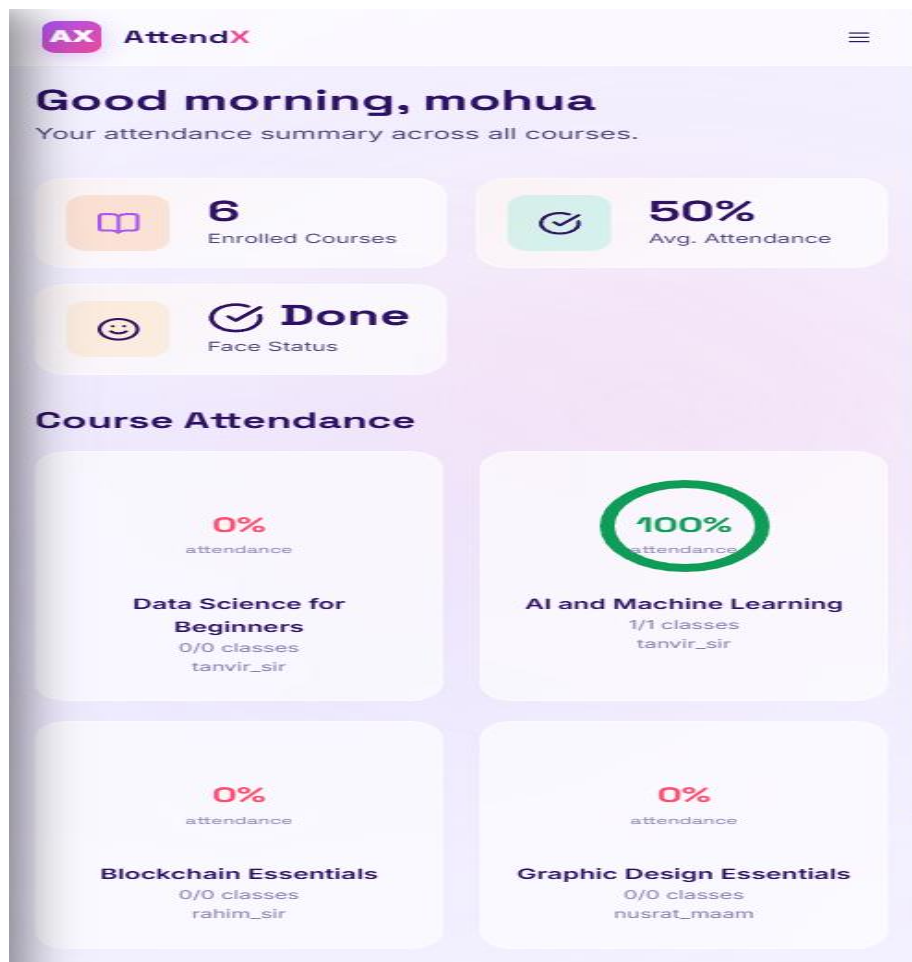


Figure 7.4 — Student Dashboard

Enrolling Your Face

Step 1: Open “Enroll,” — grant camera access and position your face within the frame.

Step 2: Capture — the system extracts and stores your facial embedding securely.

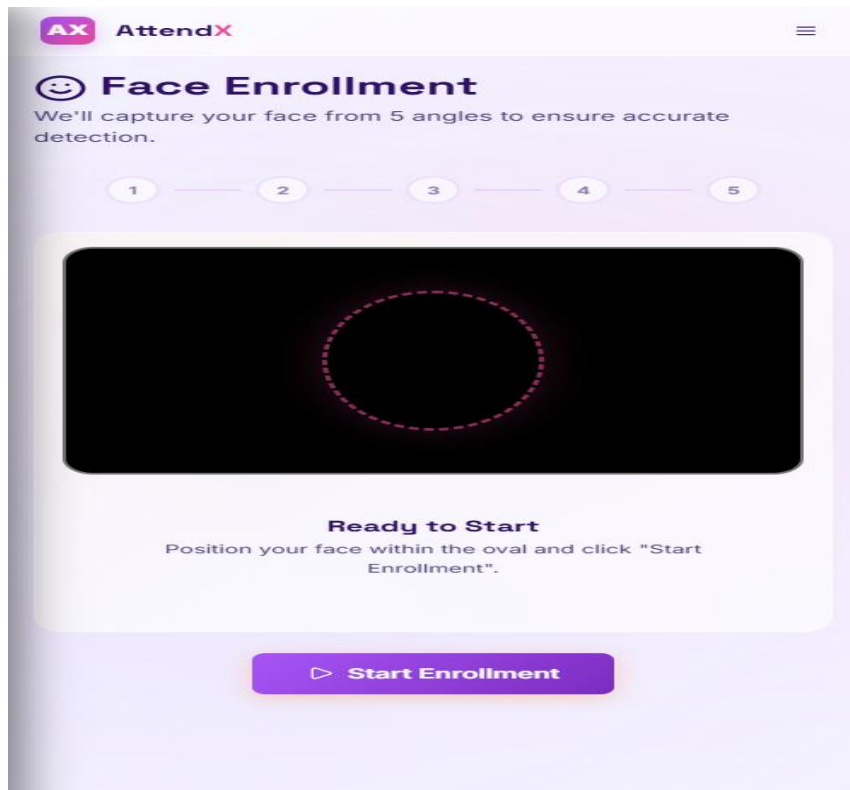


Figure 7.5 — Face Enrollment

Marking Attendance

Step 1: Open “Mark Attendance” — during an active class session.

Step 2: Look at the camera — the system matches your face against your stored embedding and confirms your attendance.

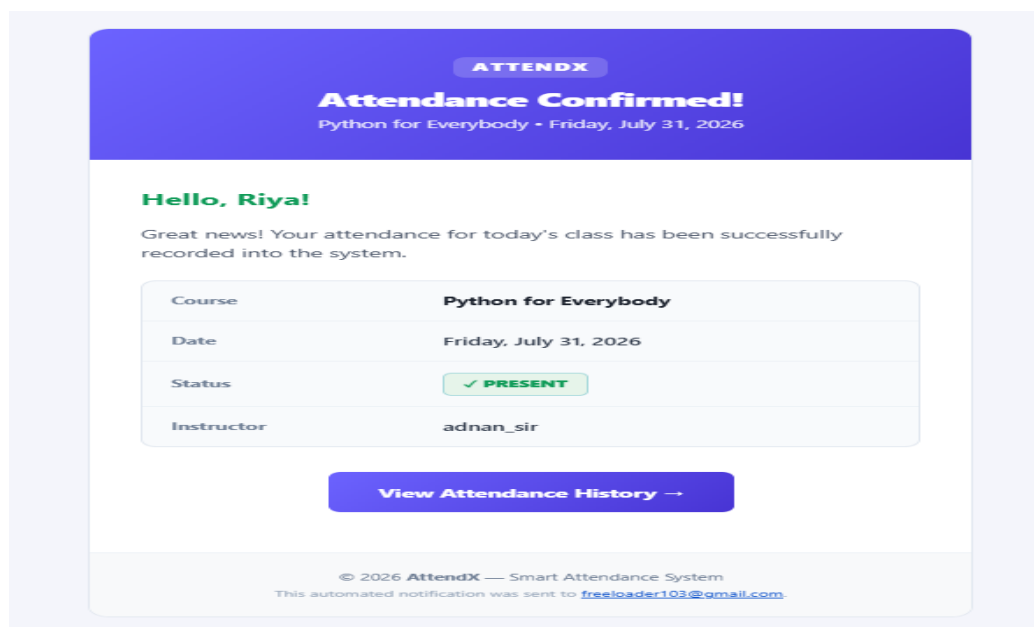


Figure 7.6 — Marking Attendance

Profile & Navigation

Use the sidebar to move between dashboard, courses, and attendance history; update your details from the profile page.

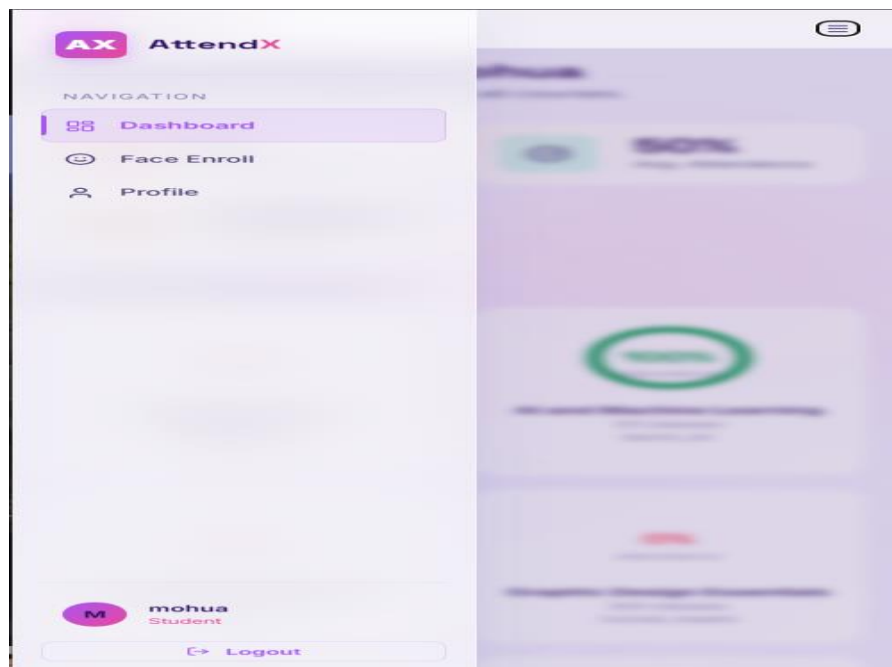


Figure 7.7 — Sidebar Navigation

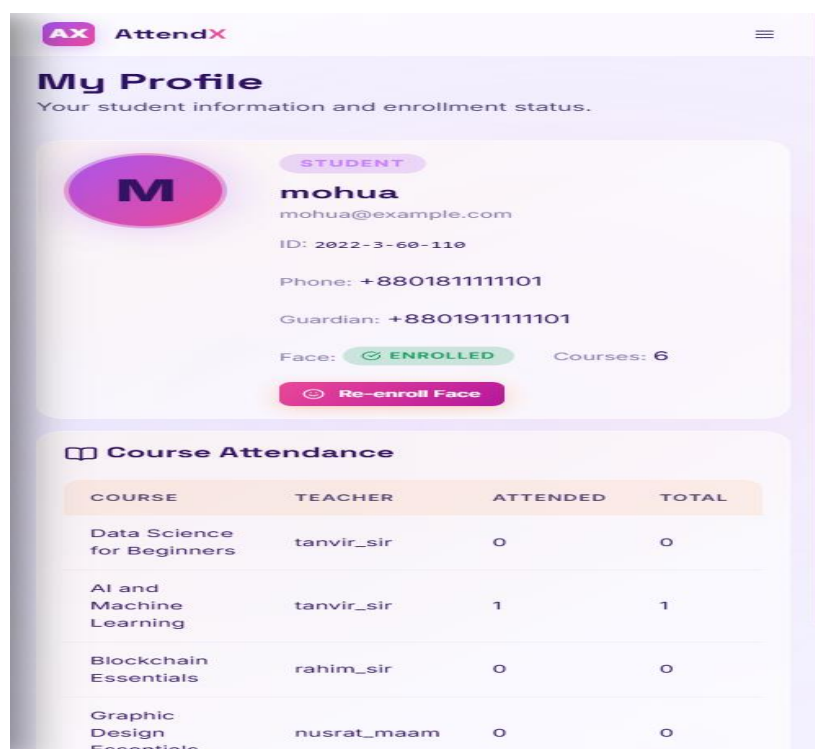


Figure 7.8 — Student Profile

7.3 Teacher Manual

View a summary of your active courses, total students, and session status.



Figure 7.9 — Teacher Dashboard

Creating & Managing Courses

- Step 1: Click “Create Course,” — enter a course code, title, and description.
- Step 2: Open a course — to view enrolled students and manage sessions.

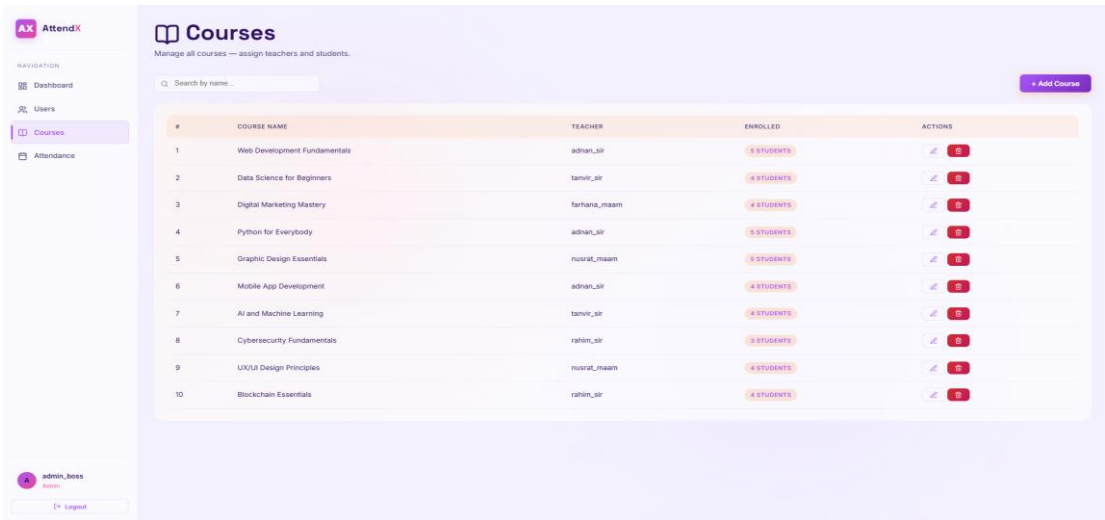


Figure 7.10 — Course Management

Launching an Attendance Session

Step 1: Open a course — and click “Start Session” to begin a time-bound attendance window, optionally with a QR code.

Step 2: Monitor live check-ins — as students verify their identity in real time.

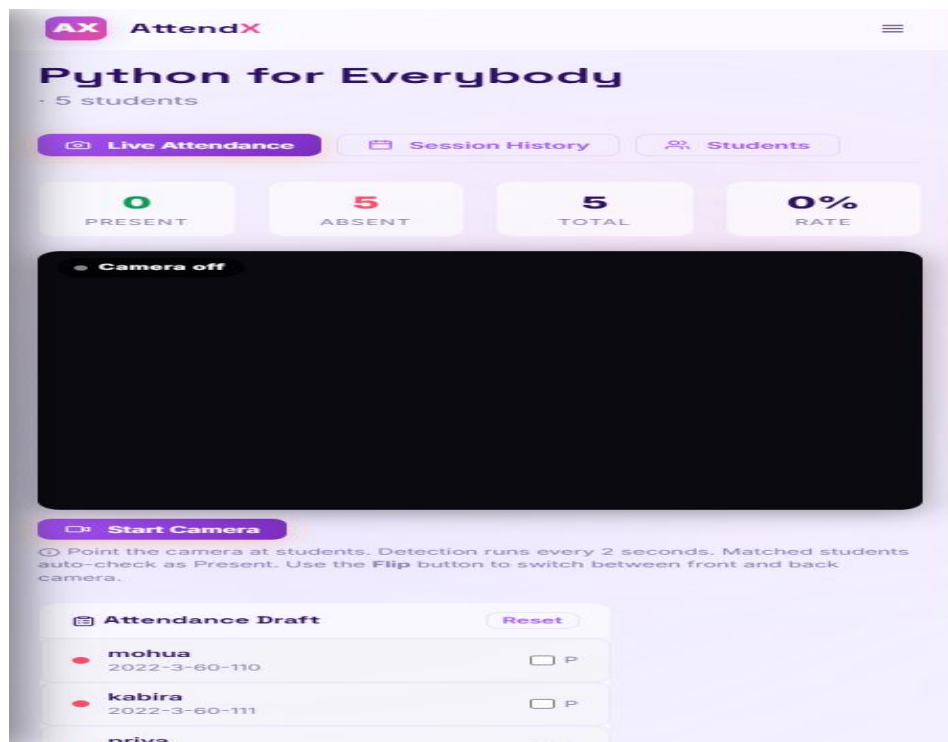


Figure 7.11 — Launching a Session

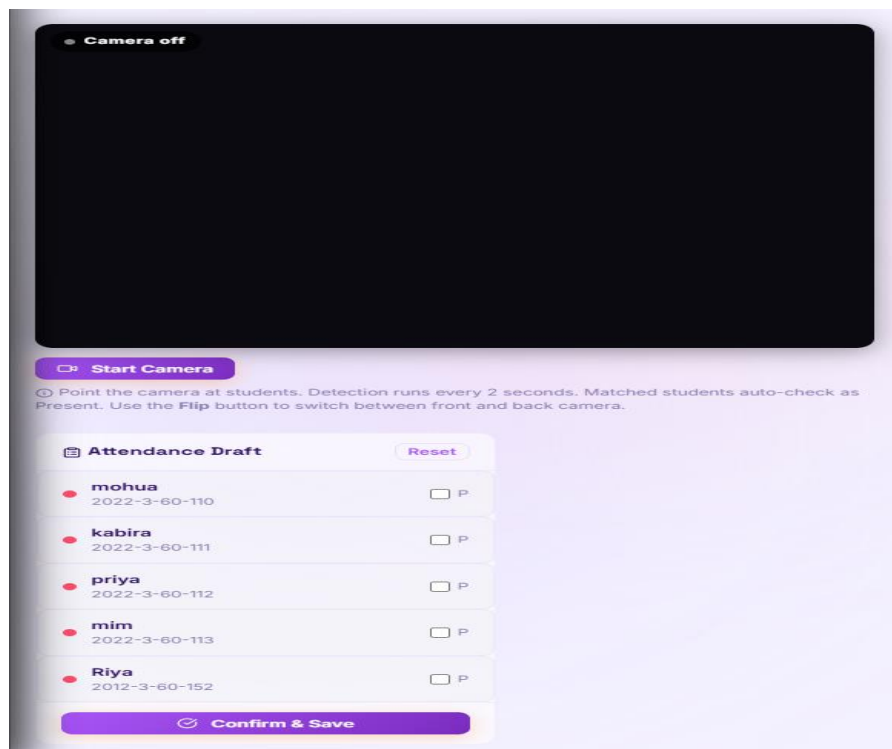
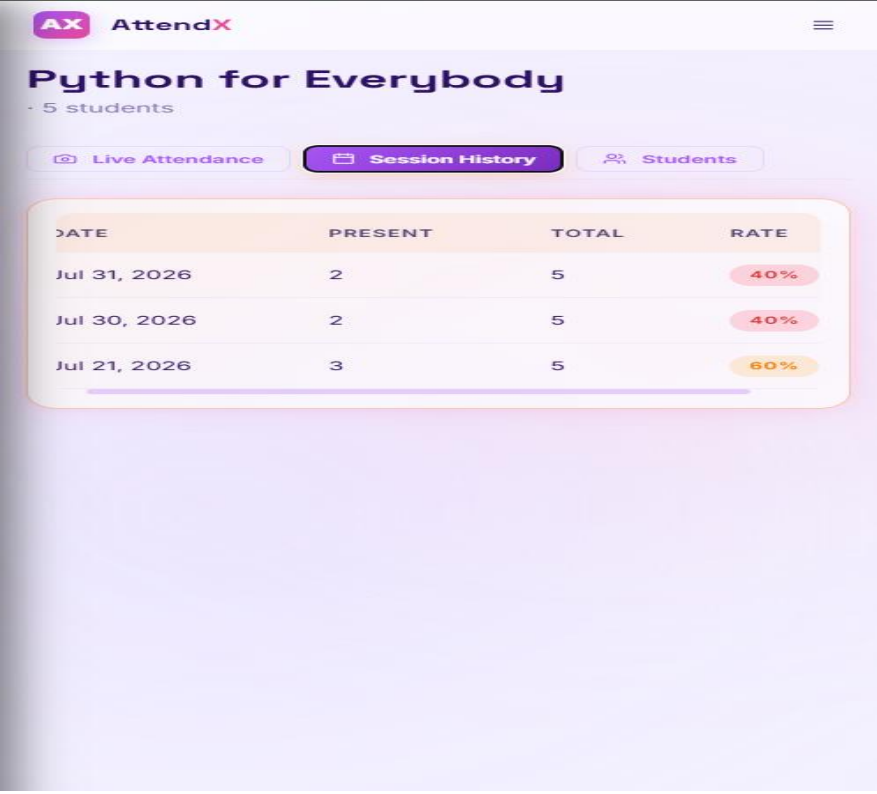


Figure 7.12 — Live Attendance Monitoring

Reports & Student Data

Review attendance history and export CSV reports; browse per-student data and flag chronic absences.



AX AttendX

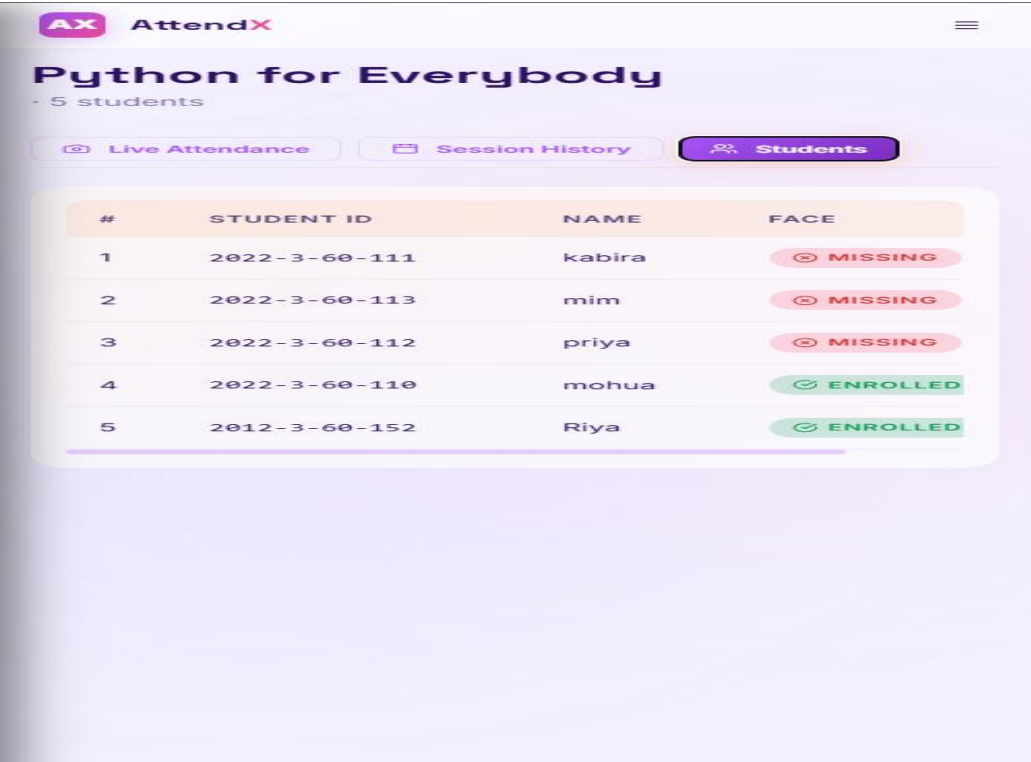
Python for Everybody

· 5 students

[Live Attendance](#) [Session History](#) [Students](#)

DATE	PRESENT	TOTAL	RATE
Jul 31, 2026	2	5	40%
Jul 30, 2026	2	5	40%
Jul 21, 2026	3	5	60%

Figure 7.13 — Attendance History



AX AttendX

Python for Everybody

· 5 students

[Live Attendance](#) [Session History](#) [Students](#)

#	STUDENT ID	NAME	FACE
1	2022-3-60-111	kabira	MISSING
2	2022-3-60-113	mim	MISSING
3	2022-3-60-112	priya	MISSING
4	2022-3-60-110	mohua	ENROLLED
5	2012-3-60-152	Riya	ENROLLED

Figure 7.14 — Student Data

7.4 Administrator Manual

The admin dashboard surfaces system-wide statistics: total users, active courses, and daily attendance volume.

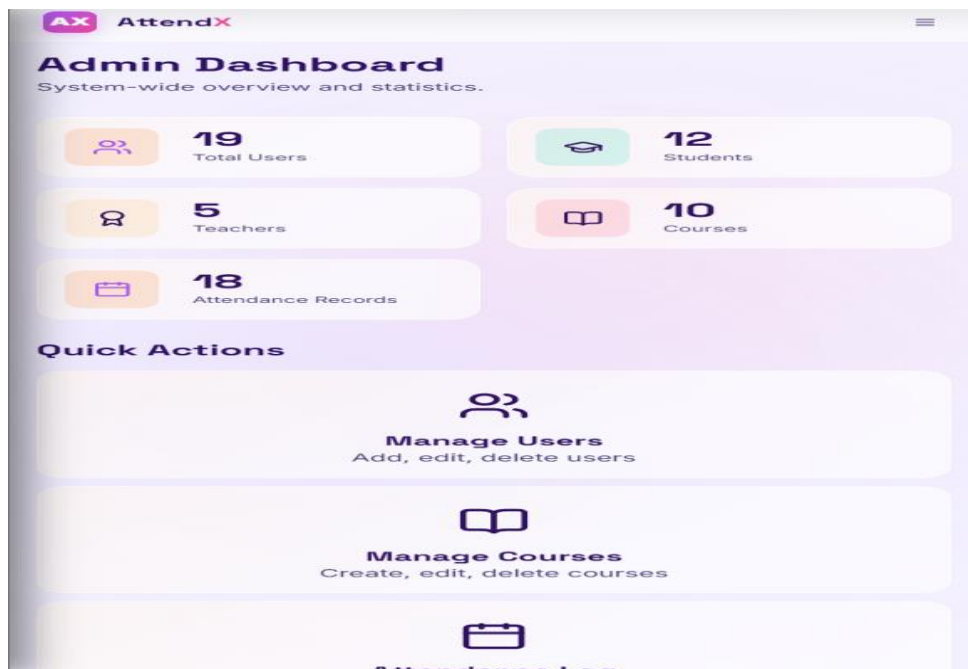


Figure 7.15 — Admin Dashboard

User Management

Search, filter, and manage student, teacher, and administrator accounts, including role changes and account status.

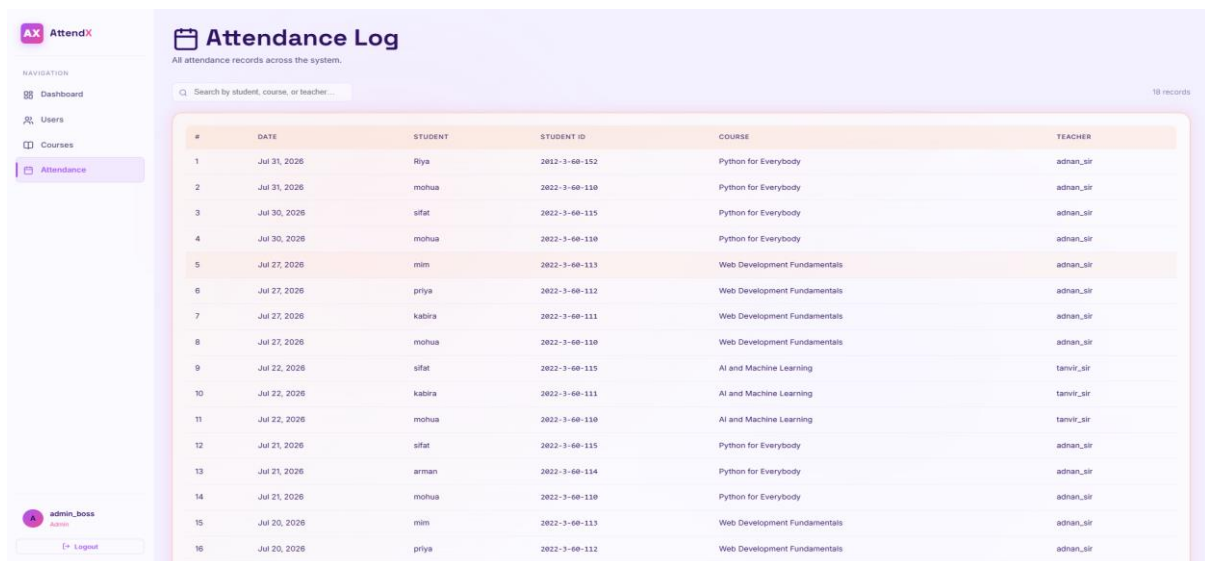
The User Management interface allows you to manage all registered users in the system. It includes a search bar and a list of users with columns for ID, Name, Email, Type, Student ID, Phone, Guardian, and Actions. A sidebar on the left shows navigation options: Dashboard, Users, Courses, and Attendance. The bottom left shows the current user's profile (admin_boss) and a Logout button.

#	NAME	EMAIL	TYPE	STUDENT ID	PHONE	GUARDIAN	ACTIONS
1	Riya	freelancer103@gmail.com	STUDENT	2012-3-60-152	+8801647336661	—	
2	System Admin	admin@attendx.com	ADMIN	—	—	—	
3	sinthia_sara	sinthia@example.com	STUDENT	2022-3-60-567	+8801716575767	+8801716573367	
4	farhana_maam	farhana@example.com	TEACHER	—	+8801711111113	—	
5	rahim_sir	rahim@example.com	TEACHER	—	+8801711111114	—	
6	nusrat_maam	nusrat@example.com	TEACHER	—	+8801711111115	—	
7	kabira	kabira@example.com	STUDENT	2022-3-60-111	+8801811111102	+8801911111102	
8	mim	mim@example.com	STUDENT	2022-3-60-113	+8801811111104	+8801911111104	
9	arman	arman@example.com	STUDENT	2022-3-60-114	+8801811111105	+8801911111105	
10	admin_boss	admin@example.com	ADMIN	—	+8801700000000	—	
11	tanila	tanila@example.com	STUDENT	2022-3-60-116	+8801811111107	+8801911111107	
12	fardin	fardin@example.com	STUDENT	2022-3-60-117	+8801811111108	+8801911111108	
13	sadia	sadia@example.com	STUDENT	2022-3-60-118	+8801811111109	+8801911111109	
14	nabil	nabil@example.com	STUDENT	2022-3-60-119	+8801811111110	+8801911111110	

Figure 7.16 — User Management

Attendance Oversight

Review attendance records across all courses and audit biometric enrollment activity for security purposes.



The screenshot displays the 'Attendance Log' interface. On the left is a navigation sidebar with 'AttendX' at the top, followed by 'NAVIGATION' and links to 'Dashboard', 'Users', 'Courses', and 'Attendance' (which is highlighted). At the bottom of the sidebar is a user profile for 'admin_boss' and a 'Logout' button. The main content area is titled 'Attendance Log' with the subtitle 'All attendance records across the system.' and a search bar 'Search by student, course, or teacher...'. A '18 records' indicator is in the top right. The table below contains 18 rows of attendance data.

#	DATE	STUDENT	STUDENT ID	COURSE	TEACHER
1	Jul 31, 2026	Riya	2022-3-60-152	Python for Everybody	adnan_sir
2	Jul 31, 2026	mohua	2022-3-60-110	Python for Everybody	adnan_sir
3	Jul 30, 2026	sifat	2022-3-60-115	Python for Everybody	adnan_sir
4	Jul 30, 2026	mohua	2022-3-60-110	Python for Everybody	adnan_sir
5	Jul 27, 2026	mim	2022-3-60-113	Web Development Fundamentals	adnan_sir
6	Jul 27, 2026	priya	2022-3-60-112	Web Development Fundamentals	adnan_sir
7	Jul 27, 2026	kabira	2022-3-60-111	Web Development Fundamentals	adnan_sir
8	Jul 27, 2026	mohua	2022-3-60-110	Web Development Fundamentals	adnan_sir
9	Jul 22, 2026	sifat	2022-3-60-115	AI and Machine Learning	tamir_sir
10	Jul 22, 2026	kabira	2022-3-60-111	AI and Machine Learning	tamir_sir
11	Jul 22, 2026	mohua	2022-3-60-110	AI and Machine Learning	tamir_sir
12	Jul 21, 2026	sifat	2022-3-60-115	Python for Everybody	adnan_sir
13	Jul 21, 2026	arman	2022-3-60-114	Python for Everybody	adnan_sir
14	Jul 21, 2026	mohua	2022-3-60-110	Python for Everybody	adnan_sir
15	Jul 20, 2026	mim	2022-3-60-113	Web Development Fundamentals	adnan_sir
16	Jul 20, 2026	priya	2022-3-60-112	Web Development Fundamentals	adnan_sir

Figure 7.17 — Attendance Overview